Ex No-9
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Implementation Of Subnetting In Cisco Packet Tracer Simulator

Aim: To implement classless IP subnetting and configure a subnetting network using Cisco packet tracer, and to verify connectivity between the devices.

Requirements:
* Cisco packet tracer
* Routers
* Switches
* PC
* Ethernet cables.

Theory: Classless IP subnetting is a technique used to divide a large IP network into smaller subnetworks called subnet. It allows the use of subnet masks other than the default classful masks, thereby making efficient use of IP addresses. The network 192.168.1.0/24 is divided using a /27 subnet mask. The /27 mask provides 8 subnets, with 30 usable host addresses in each subnet. Subnetting helps in reducing IP address wastage, controlling broadcast traffic, improving network performance and simplified network management.

Network Topology: A network topology consisting of routers, switches and PCs is created in Cisco Packet Tracer. The devices are connected using appropriate Ethernet cables. Each group of PC's is connected to a switch, and the switches are connected to the router interfaces.

IP addressing: The IP addresses are assigned according to the different subnets.

Router R1: * Gigabit Ethernet 0/0 - 192.168.1.1/27
* Gigabit Ethernet 0/1 - 192.168.2.1/27

Network 1 - 192.168.1.0/27:

* PC1 - 192.168.1.11
* PC2 - 192.168.1.12
* PC3 - 192.168.1.13
* PC4 - 192.168.1.14
* PC5 - 192.168.1.15

Network 2 - 192.168.2.0/27:

* PC1 - 192.168.2.11
* PC2 - 192.168.2.12
* PC3 - 192.168.2.13
* PC4 - 192.168.2.14
* PC5 - 192.168.2.15

Router R2:

* Fast Ethernet 0/0 - 192.168.3.1/27
* fast Ethernet 0/1 - 192.168.4.1/27

Network 3 - 192.168.3.0/27

* PC1 = 192.168.3.11
* PC2 - 192.168.3.12
* PC3 - 192.168.3.13
* PC4 - 192.168.3.14
* PC5 - 192.168.3.15

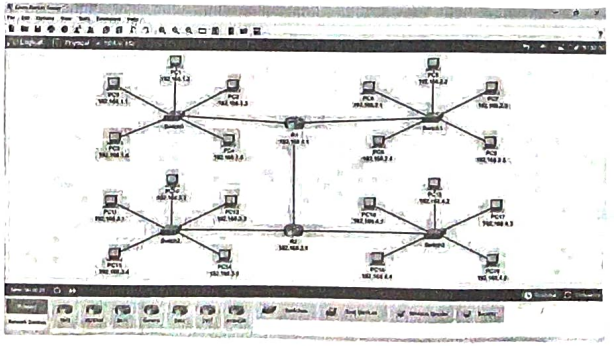
Network 4 - 192.168.4.0/27

* PC1 - 192.168.4.11
* PC2 - 192.168.4.12
* PC3 - 192.168.4.13
* PC4 - 192.168.4.14
* PC5 - 192.168.4.15

PROCEDURE:

1) Open Cisco packet tracer and create a new network topology.
2) Add the required routers, switches and PCs to the workspace.
3) Connect the devices using suitable Ethernet cables.
4) Divide the network into smaller subnets using the /27 subnet mask.
5) Configure the router interfaces with the appropriate IP addresses and subnet mask.
6) Enable the router interfaces using the no shutdown command.
7) Configure the switch ports in access mode.
8) Assign the appropriate IP address and subnet mask to each PC.
9) Configure the default gateway of each PC as the IP address of the corresponding router interface.
10) Check the connectivity between PCs and routers using the ping command.
11) Verify the packets are successfully transmitted between the devices.

OBSERVATION: The 192.168.1.0/24 network was successfully divided into smaller /27 subnets. Each subnet provides 30 usable host addresses. The routers, switches and PCs were assigned IP addresses according to their respective subnets. The router interfaces were enabled and the PC's were configured with suitable default gateways. Successfully ping responses confirmed connectivity between the devices.



RESULT: Thus classless IP subnetting was successfully implements in Cisco packet Tracer by dividing the network into small /27 subnets, configured the required devi and verified communication between the connected devices.